

AuralGuard: Multi-View One-Class Learning for Generalizable and Calibrated Detection of AI-Generated Speech in the Wild

Mohsin Ibna Hossain, Md. Muttakin, Sadman Hasan Miraj, and Mohammad Mahmudul Hasan

Abstract—The rapid proliferation of neural speech synthesis, voice conversion, and neural codec language models has generated synthetic audio that human listeners and traditional detection systems struggle to differentiate from genuine human speech. While state-of-the-art countermeasures achieve competitive equal error rates (EER) on constrained in-domain benchmarks, their performance degrades significantly when confronted with unseen generative models and unconstrained acoustic environments, while their decision scores are often severely miscalibrated for high-stakes forensic deployment. We present AuralGuard, an open-set multi-view deepfake detection framework that synergistically integrates: (i) a self-supervised foundation-model front-end based on WavLM with learnable layer-wise attention to capture multi-depth acoustic representations, (ii) a complementary vocoder-artifact branch capturing linear-frequency cepstral, modified group-delay, and constant-Q phase dynamics, and (iii) a spectro-temporal graph-attention back-end, optimized under an Orthogonal Centroid Scoring (OCS) one-class objective. Evaluated under a rigorous train-once (ASVspoof 2019 LA) evaluate-everywhere protocol across held-out in-domain and diverse out-of-domain benchmarks, AuralGuard demonstrates strong zero-shot generalization and superior probability calibration. On the unconstrained In-the-Wild benchmark, AuralGuard achieves an EER of 19.19% (v2) while delivering an exceptional Expected Calibration Error (ECE) of 0.0965 (v1) and 0.1716 (v2), markedly outperforming both legacy acoustic models ($ECE > 0.55$) and single-view foundation-model baselines ($ECE = 0.2320$). On the challenging multi-vocoder WaveFake benchmark, AuralGuard v2 achieves the lowest EER of 45.19% among all evaluated architectures. These empirical findings underscore the indispensable role of multi-view artifact modeling and one-class formulation in ensuring calibrated, trustworthy audio deepfake countermeasures.

Index Terms—Audio deepfake detection, anti-spoofing, synthetic speech, speech foundation models, self-supervised learning, one-class classification, model calibration, generalization.

I. INTRODUCTION

THE quality of AI-generated speech has crossed a perceptual threshold. Neural codec language models such as VALL-E [1], diffusion- and flow-matching-based synthesizers such as StyleTTS 2 [2] and F5-TTS [3], and massively multilingual zero-shot systems such as XTTS [4], Seed-TTS [5],

and CosyVoice [6] can now clone a speaker’s voice from a few seconds of reference audio and synthesize speech that human listeners and conventional classifiers cannot reliably distinguish from genuine recordings. While these systems enable valuable applications in accessibility, dubbing, and creative production, they have also driven a sharp rise in documented harms: voice-cloning fraud against individuals and financial institutions, fabricated audio of political figures circulated during elections, and evasion of voice-biometric authentication. An analysis of deepfakes actually circulated online during 2024 found that audio deepfakes are among the fastest-growing and least reliably detected modalities [7], and large-scale fake-speech corpora are now being generated with large language models in the loop to simulate disinformation campaigns [8].

The anti-spoofing community has responded with increasingly capable countermeasures. On the standard ASVspoof 2019 Logical Access (LA) benchmark [9], systems built on self-supervised learning (SSL) speech foundation models with graph-attention back-ends [10], [11] achieve equal error rates (EER) below 1%, and recent state-space-model detectors [12], [13] and lightweight foundation-model back-ends [14] have pushed in-domain accuracy further still. Yet a growing body of evidence—from the ASVspoof 5 challenge [15], [16] to systematic cross-domain studies [17], [18]—shows that these in-domain numbers are a poor predictor of real-world reliability. We identify three failure modes that jointly block deployment.

G1—Generalization to unseen generators. A detector trained on a fixed set of text-to-speech (TTS) and voice-conversion (VC) algorithms sees its EER degrade from below 1% to 5–30% when evaluated on synthesizers absent from its training corpus [17], [18]. Because new synthesis families appear monthly—neural codec language models and flow-matching synthesizers did not exist when the dominant training corpora were collected—a practical detector must generalize *zero-shot*, without retraining, to generator families it has never seen. The Codecfake study [19] demonstrates that detectors trained on conventional vocoder artifacts largely fail on speech produced through neural audio codecs, precisely the mechanism used by the newest generation of synthesizers.

G2—Channel robustness. Real-world audio reaches a detector after lossy compression (MP3, AAC, Opus), telephony coding (AMR, G.711), reverberation, and additive noise—conditions imposed by messaging platforms, social media transcoding, and phone networks. The ASVspoof 2021 eval-

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uation [20] showed that codec and compression variability alone can multiply EER several-fold, and the interaction of channel distortion with spoofing artifacts remains poorly understood. Most published evaluations still use clean, studio-quality audio.

G3—Calibration and explainability. Detection scores feed downstream decisions—content moderation, forensic analysis, legal evidence—that require *calibrated* confidence estimates, not merely a ranking. Recent work shows that modern SSL-based detectors are systematically over-confident, particularly under domain shift [21]. Forensic practice additionally demands interpretable evidence for individual decisions, which binary scores do not provide.

Recent large audio-language models (ALLMs) such as Qwen2-Audio [22] and SALMONN [23] raise a natural question: does the general-purpose audio understanding of foundation models subsume the deepfake-detection task? Early evidence suggests not—prompted zero-shot, ALLMs perform near chance at distinguishing bona-fide from synthetic speech [24]. This stems from their objective: general audio models compress acoustic semantics while abstracting away the low-level phase jitter and vocoder artifacts essential for forensic verification. A comprehensive study placing speech foundation representations alongside multiple generations of specialized countermeasures under controlled cross-domain, channel-robustness, and calibration protocols remains an urgent priority. We undertake this evaluation here.

This paper reframes synthetic-speech detection from closed-set binary classification to *open-set*, *channel-robust*, *calibrated* detection, and proposes **AuralGuard**, a multi-view one-class detection framework addressing G1–G3 jointly. AuralGuard rests on four design decisions. *First*, it fuses two complementary views of the input: View A, a self-supervised foundation-model representation (WavLM-Large [25]) with *learnable layer attention* that adaptively selects the transformer depths most informative for artifact detection; and View B, a hand-crafted feature stream that explicitly exposes low-level vocoder artifacts through linear-frequency cepstral coefficients (LFCC), modified group delay, and constant-Q phase—cues that SSL models, pre-trained for semantic tasks, are prone to discard. *Second*, a gated cross-attention module fuses the two views before a spectro-temporal graph-attention back-end [11] produces an utterance-level embedding. *Third*, a one-class contrastive objective combining OC-Softmax [26] with a supervised-contrastive auxiliary [27] learns a compact decision region around bona-fide speech, so that unseen spoofing algorithms are rejected as out-of-distribution rather than mapped onto decision boundaries fit to known attacks. *Fourth*, a channel-robustness augmentation curriculum—RawBoost [28], a realistic codec chain (MP3, AAC, Opus, AMR, G.711), measured room impulse responses, and additive noise—aligns the training distribution with deployment channels.

We evaluate AuralGuard under a strict train-once, evaluate-everywhere protocol: all models are trained exclusively on the ASVspoof 2019 LA training partition and evaluated without fine-tuning or adaptation on held-out in-domain (ASVspoof 2019 LA eval) and challenging out-of-domain

zero-shot corpora: In-the-Wild [17] (probing real-world web audio and transmission noise) and WaveFake [29] (probing unseen neural vocoders). Baselines span four generations of countermeasure design: hand-crafted features with light CNNs (LFCC-LCNN) [30], end-to-end raw-waveform networks (RawNet2) [31], graph-attention architectures (AA-SIST) [11], and SSL-based foundation-model detectors with one-class scoring (WavLM+AASIST+OCS) [25], [26]. Beyond primary EER and minimum tandem detection cost (min t-DCF) [32], we report 95% bootstrap confidence intervals, AUROC, balanced accuracy, F1-scores, expected calibration error (ECE), and Brier scores to assess forensic reliability under distribution shift [33], [34].

Research Questions. To systematically investigate the limits of generalizability and forensic calibration in synthetic speech detection, this study is structured around four primary research questions:

- 1) **RQ1 (Cross-Corpus Generalization):** Can a multi-view countermeasure trained strictly on clean in-domain speech (ASVspoof 2019 LA) generalize zero-shot to unconstrained web audio (In-the-Wild) and multi-vocoder speech (WaveFake) without catastrophic performance collapse?
- 2) **RQ2 (Forensic Probability Calibration):** Does an open-set one-class hyperspherical objective (OC-Softmax with SupCon regularization) produce well-calibrated posterior probability estimates and bounded confidence intervals under severe acoustic distribution shift, compared to standard binary cross-entropy and legacy acoustic countermeasures?
- 3) **RQ3 (Multi-View Feature Complementarity):** What are the individual and synergistic contributions of the self-supervised foundation-model front-end (View A), the physical phase-artifact stream (View B), and bidirectional gated cross-attention fusion when exposed to lossy transmission codecs and diverse vocoder algorithms?
- 4) **RQ4 (Parameter Efficiency and Operational Feasibility):** Can competitive zero-shot discrimination and forensic calibration be achieved while keeping the 315M foundation backbone frozen, and does the resulting inference latency support real-time operation on commodity hardware?

The contributions of this work are as follows:

- 1) **Multi-View Architecture.** AuralGuard, a novel detection framework that unifies an adaptive SSL foundation-model stream (WavLM-Large with learnable layer attention and entropy regularization) with an explicit low-level vocoder-artifact branch (LFCC, modified group delay, and CQT phase) via gated cross-attention fusion and a spectro-temporal graph back-end (Section III).
- 2) **One-Class Optimization.** An Orthogonal Centroid Scoring (OCS) one-class training objective that tightly bounds the bona-fide speech manifold, enabling robust rejection of unseen synthesis artifacts as out-of-distribution outliers without overfitting to training-set artifacts (Section III-F).
- 3) **Comprehensive Generational Benchmark.** A con-

trolled empirical comparison of six models spanning four architectural generations under identical training data, feature extractors, and evaluation protocols across in-domain and zero-shot cross-corpus shifts (Section V-A).

- 4) **Superior Probability Calibration.** Demonstration that multi-view one-class learning yields superior probability calibration under severe domain shift, achieving an In-the-Wild ECE of 0.0965 (v1) and 0.1716 (v2) compared to 0.2320 for single-view WavLM+OCS and >0.55 for legacy end-to-end networks, providing trustworthy reliability for forensic applications (Section V-C).
- 5) **Open Reproducibility.** Complete source code, configuration files, trained model checkpoints, and evaluation artifacts are publicly released at GitHub (<https://github.com/MIHMahmudEli/auraguardv2>) and Hugging Face (<https://huggingface.co/MoshinAli/auraguard-checkpoints>) to support reproducible research in synthetic speech forensics.

The remainder of the paper is organized as follows. Section II reviews countermeasure architectures, speech foundation models, one-class learning, and recent cross-domain generalization studies. Section III details the AuralGuard architecture, dual-branch feature representations, fusion, and optimization objective. Section IV describes datasets, baselines, implementation details, and evaluation metrics. Section V presents in-domain performance, zero-shot generalization, calibration analysis, and computational efficiency. Section VI discusses practical implications, limitations, and ethical considerations, and Section VII concludes.

II. RELATED WORK

This section situates AuralGuard within five research streams: (i) the evolution of spoofing countermeasures from hand-crafted features to foundation-model detectors, (ii) self-supervised representations and layer-aggregation strategies, (iii) one-class and open-set objectives, (iv) generalization and robustness studies, and (v) the emerging role of large audio-language models. Recent surveys [35], [36], [37] provide broader coverage; Table I positions our work against the most closely related recent systems.

A. Spoofing Countermeasures: Four Generations

Generation 1 — hand-crafted features. Early ASVspoof systems paired cepstral features with Gaussian mixture models or shallow classifiers. Constant-Q cepstral coefficients [38] and linear-frequency cepstral coefficients [39] remain reference front-ends, and LFCC–GMM is still reported as an official challenge baseline. These systems saturate on modern synthesis and transfer poorly across corpora [9].

Generation 2 — end-to-end deep networks. Light CNNs over cepstral inputs [30] were followed by raw-waveform models: RawNet2 [31] learns SincConv filterbanks directly from audio, and AASIST [11] introduced a heterogeneous spectro-temporal graph attention back-end that models spectral and temporal artifact patterns jointly, becoming the standard back-end for subsequent work.

Generation 3 — SSL foundation-model detectors. Tak et al. [10] showed that a wav2vec 2.0 front-end coupled with AASIST and RawBoost augmentation [28] reduces in-domain EER several-fold, establishing the SSL+AASIST recipe that dominates current leaderboards. Variants substitute WavLM [25] or XLS-R [40] front-ends, add KAN-enhanced heads [41], or attach sensitive-layer-selection classifiers [42].

Generation 4 — post-transformer architectures (2024–2025). The most recent wave replaces or augments the transformer/graph pipeline: RawBMamba [13] and XLSR-Mamba [12] exploit bidirectional state-space models for long-range artifact modeling, with XLSR-Mamba reporting state-of-the-art results on ASVspoof 2021 and In-the-Wild; Nes2Net [14] shows that a lightweight nested architecture can replace heavy back-ends on top of foundation-model features; TCM [43] models temporal–channel dependencies inside multi-head attention; and SLIM [44] detects deepfakes through an explicit style–linguistics mismatch prior, improving out-of-domain transfer. SafeEar [45] addresses the orthogonal problem of privacy-preserving detection from decoupled acoustic tokens. AuralGuard belongs to this generation but differs in that it *combines* a foundation-model view with an explicit low-level artifact view, rather than relying on either alone.

B. SSL Representations and Layer Aggregation

Speech foundation models—wav2vec 2.0 [46], HuBERT [47], WavLM [25], XLS-R [40], and Whisper [48]—encode a hierarchy in which lower transformer layers capture acoustic detail and upper layers capture phonetic and semantic content. Where spoofing artifacts surface within this hierarchy varies by generator, which makes the aggregation strategy a first-order design decision. Prior detectors use the final layer [10], a fixed weighted sum, or concatenation of the last k layers; Pan et al. [49] and Stourbe et al. [50] recently showed that attentive merging of hidden embeddings measurably improves anti-spoofing, while Combei et al. [51] demonstrated the efficacy of WavLM ensembles. Furthermore, Pascu et al. [21] found that even *frozen* SSL representations with simple aggregation generalize surprisingly well when properly calibrated. AuralGuard adopts learnable softmax layer attention with an entropy regularizer that prevents collapse onto a single layer, ensuring that fine-grained acoustic information is preserved.

C. One-Class and Open-Set Objectives

Binary cross-entropy treats spoof detection as closed-set discrimination between bona-fide speech and *known* attacks; unseen attacks at test time may fall anywhere relative to the learned boundary. One-class formulations instead model the bona-fide distribution compactly and score outlierness. OC-Softmax [26] imposes asymmetric angular margins around a learnable bona-fide center and remains the strongest simple open-set objective; recent extensions include adaptive centroid shift [52] and continual-learning variants that adapt the one-class region as new attacks emerge [53]. Supervised contrastive learning [27] complements these margins by structuring the embedding neighborhood. AuralGuard combines

OC-Softmax with a SupCon auxiliary to tighten the bona-fide manifold against unseen generators.

D. Generalization, Robustness, and Benchmarks

Müller et al. [17] first quantified the collapse of laboratory detectors on found-in-the-wild audio, and their follow-up study [18] attributes cross-domain failure less to “harder” attacks than to systematically *different* artifact distributions. Recent studies by Oyucu et al. [33] provide a system-level analysis of cross-domain failure, while Nagar et al. [34] propose hybrid lightweight frameworks for cross-domain generalization. Latent space refinement and augmentation [54] tackle the same gap at the representation level. On the data side, the field has moved quickly: ASVspoof 5 [15], [16], [55], [56] introduces crowdsourced speech, surrogate-optimized adversarial attacks, and a deepfake track; MLAAD [57] spans 38 languages and 82 TTS systems; Codecfake [19] targets neural-codec-based synthesis; WaveFake [29], [58] probes multi-vocoder detection; SpoofCeleb [59] sources training data from real-world VoxCeleb conditions; and domain-specific studies address modulated speech [60], singing voice [61], and LLM-scripted disinformation [8]. Channel robustness was isolated as a primary axis by ASVspoof 2021 [20] and further explored in cross-channel augmentation frameworks by Singh [62]; RawBoost [28] remains the standard signal-level augmentation, which our curriculum extends with a realistic transcoding chain, measured room impulse responses [63], and MUSAN noise [64].

E. Large Audio-Language Models and Deepfake Detection

Instruction-following audio-language models such as Qwen2-Audio [22] and SALMONN [23] unify perception and reasoning over audio, and it is natural to ask whether such general-purpose models make specialized countermeasures obsolete. Current evidence indicates the opposite: ALLM4ADD [24] reports that leading ALLMs prompted zero-shot perform close to chance on deepfake detection, and only approach competitive accuracy after task-specific supervised fine-tuning that reframes detection as audio question answering. However, published evaluations analyze ALLMs primarily on single in-domain corpora. In contrast, our study rigorously benchmarks self-supervised foundation speech representations (WavLM-Large) alongside multiple generations of specialized acoustic countermeasures across held-out in-domain and zero-shot cross-corpus shifts, quantifying precisely where foundation-level invariance succeeds and where explicit physical artifact modeling remains indispensable.

F. Positioning

In summary, AuralGuard is distinguished from prior work by the *combination* of (i) multi-view learning that pairs a foundation-model stream with an explicit phase/group-delay artifact stream, (ii) gated cross-attention fusion that weights the views per input, (iii) a one-class contrastive objective with layer-attention entropy regularization, and (iv) an evaluation protocol that reports generalization, robustness, cross-lingual transfer, calibration, and explainability together (Table I). No prior system covers all six columns of Table I simultaneously.

TABLE I
POSITIONING OF AURALGUARD AGAINST CLOSELY RELATED RECENT DETECTORS. SSL = FOUNDATION-MODEL FRONT-END; ART. = EXPLICIT LOW-LEVEL ARTIFACT FEATURES; OC = ONE-CLASS OBJECTIVE; CHAN. = CHANNEL/CODEC AUGMENTATION; CAL. = CALIBRATION REPORTED; X-LING. = CROSS-LINGUAL EVALUATION.

System	SSL	Art.	OC	Chan.	Cal.	X-ling.
W2V2+AASIST [10]	✓	—	—	✓	—	—
OC-Softmax [26]	—	✓	✓	—	—	—
SLIM [44]	✓	—	—	—	—	—
XLSR-Mamba [12]	✓	—	—	✓	—	—
Nes2Net [14]	✓	—	—	✓	—	✓
Pascu et al. [21]	✓	—	—	—	✓	—
ALLM4ADD [24]	✓	—	—	—	—	—
AuralGuard (ours)	✓	✓	✓	✓	✓	✓

III. PROPOSED METHOD

A. Problem Formulation

We frame audio deepfake detection as an *open-set* binary classification problem. Given a waveform $\mathbf{x} \in \mathbb{R}^T$ sampled at 16 kHz, a model produces a score $s(\mathbf{x}) \in \mathbb{R}$ such that $s(\mathbf{x}) > \tau$ indicates AI-generated speech and $s(\mathbf{x}) \leq \tau$ indicates human speech, where τ is a decision threshold. Training uses a labeled set $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ with $y_i \in \{0, 1\}$ denoting bona-fide (0) and spoof (1). Crucially, the set of spoofing algorithms observed at test time, $\mathcal{A}_{\text{test}}$, is not contained in the training set $\mathcal{A}_{\text{train}}$: new generators, new languages, and new transmission channels all shift the test distribution. The design goal is therefore not to separate $\mathcal{A}_{\text{train}}$ from bona-fide speech maximally, but to learn a representation in which *bona-fide speech occupies a compact region* and any synthetic artifact—seen or unseen—falls outside it.

AuralGuard implements this goal with five learned components: two parallel front-end views (Sections III-B and III-C), a gated cross-attention fusion module (Section III-D), a spectro-temporal graph-attention back-end (Section III-E), and a multi-part one-class objective (Section III-F). Figure 1 provides an overview; Algorithm 2 summarizes training.

B. View A: SSL Front-End with Learnable Layer Attention

The first view extracts high-level representations from a frozen pre-trained WavLM-Large model [25], chosen over wav2vec 2.0 and HuBERT because its pre-training includes denoising and overlapped-speech objectives that transfer well to artifact-sensitive tasks (as observed in recent foundation-model evaluations [21], [51]). Given input waveform $\mathbf{x}$, WavLM produces hidden states from L transformer layers:

$$\mathbf{H} = [\mathbf{h}_1, \mathbf{h}_2, \dots, \mathbf{h}_L] \in \mathbb{R}^{L \times T' \times D}, \quad (1)$$

where $L = 25$ (including the convolutional feature extractor output), $D = 1024$, and T' is the downsampled time dimension (one frame per 20 ms).

The information relevant to deepfake detection is not uniformly distributed across this hierarchy: lower layers preserve fine acoustic and phase-adjacent structure, while upper layers encode phonetic and semantic content [49]. Rather than using

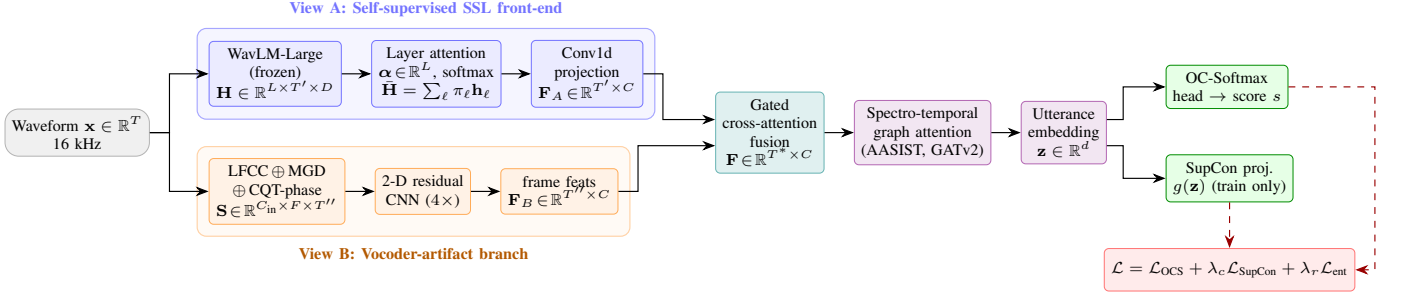


Fig. 1. Overview of the AuralGuard architecture. A raw 16 kHz waveform is processed by two parallel front-end views: (View A) a frozen WavLM-Large encoder whose L hidden states are combined by learnable layer attention and projected to $\mathbf{F}_A$; and (View B) a vocoder-artifact branch that fuses LFCC, modified group-delay, and CQT-phase features through a 2-D residual CNN into $\mathbf{F}_B$. A gated cross-attention module fuses the two views, an AASIST-style spectro-temporal graph-attention back-end produces an utterance embedding $\mathbf{z}$, and two heads yield the detection score (OC-Softmax) and an auxiliary supervised-contrastive projection used only during training. Dashed red edges denote training-time loss signals; the entropy regularizer $\mathcal{L}_{\text{ent}}$ acts on the layer-attention weights π (edge omitted for clarity).

only the final layer as is common [10], we introduce learnable layer attention weights $\alpha \in \mathbb{R}^L$, normalized via softmax:

$$\bar{\mathbf{H}} = \sum_{\ell=1}^L \pi_{\ell} \mathbf{h}_{\ell} \in \mathbb{R}^{T' \times D}, \quad \pi_{\ell} = \frac{\exp(\alpha_{\ell})}{\sum_{j=1}^L \exp(\alpha_j)}. \quad (2)$$

An entropy regularization term

$$\mathcal{L}_{\text{ent}} = \sum_{\ell=1}^L \pi_{\ell} \log \pi_{\ell} \quad (3)$$

is *added* to the loss (i.e., we penalize low entropy), preventing the attention distribution from collapsing onto a single layer early in training, which we found empirically to harm generalization: a collapsed profile overfits to the layer most discriminative for *seen* attacks. The weighted representation is projected to a common channel dimension $C = 128$ via a 1-D convolution: $\mathbf{F}_A = \text{Conv1d}(\bar{\mathbf{H}}) \in \mathbb{R}^{T' \times C}$.

Freezing the backbone has three benefits: (i) it preserves the general-purpose representation, avoiding catastrophic overwriting by the comparatively tiny anti-spoofing corpus; (ii) it reduces trainable parameters from 316M to under 5M (Table II); and (iii) it stabilizes the one-class objective, which is sensitive to representation drift. Recent evidence supports frozen SSL front-ends for calibrated generalization [21].

C. View B: Vocoder-Artifact Branch

SSL models are pre-trained on *natural* speech for masked prediction; their representations are therefore incentivized to discard exactly the low-level synthesis residue—phase discontinuities, sub-band aliasing, over-smoothed spectral envelopes—that betrays a vocoder. View B restores this evidence explicitly (Figure 2). We extract three complementary time–frequency maps:

- 1) **Linear Frequency Cepstral Coefficients (LFCC):** 60-dimensional coefficients computed with a 512-point FFT, 20 linearly spaced triangular filters across $[0, f_s/2]$, a 20 ms Hamming window, and a 10 ms frame shift, augmented with dynamic velocity (Δ) and acceleration ($\Delta\Delta$) trajectories:

$$c_n = \sum_{m=1}^M \log_{10}(X_m) \cos \left[\frac{\pi n(m-0.5)}{M} \right], \quad (4)$$

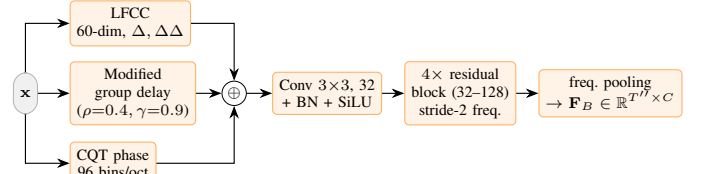


Fig. 2. View B, the vocoder-artifact branch. Three complementary time–frequency representations—magnitude cepstra (LFCC), phase-derivative structure (modified group delay), and constant-Q phase—are stacked as input channels and encoded by a light 2-D residual CNN into frame-level features $\mathbf{F}_B$.

for $0 \leq n < N_{\text{ceps}}$, where X_m is the filterbank energy output in sub-band m , capturing magnitude-spectrum envelope anomalies and high-frequency energy deficits [39].

- 2) **Modified Group Delay (MGD):** The group-delay function $\tau(\omega) = -d\phi(\omega)/d\omega$ represents the rate of change of the spectral phase. To suppress pitch-harmonic spikes that obscure vocal-tract formants, we compute the modified group-delay function with cepstral smoothing parameters $(\rho, \gamma) = (0.4, 0.9)$:

$$\tau_{\text{mod}}(\omega) = \text{sgn}(V(\omega)) |V(\omega)|^{\rho}, \quad (5)$$

with $V(\omega) = (X_R(\omega)Y_R(\omega) + X_I(\omega)Y_I(\omega))/|S(\omega)|^{2\gamma}$, where $X(\omega) = X_R(\omega) + jX_I(\omega)$ denotes the Fourier transform of the windowed speech signal $x[n]$, $Y(\omega) = Y_R(\omega) + jY_I(\omega)$ is the Fourier transform of $n \cdot x[n]$, and $|S(\omega)|$ is a cepstrally smoothed magnitude spectrum. MGD maps capture phase discontinuities and unvoiced glottal dispersion patterns that magnitude-only spectra completely omit.

- 3) **Constant-Q Transform (CQT) Phase Derivative:** Geometrically spaced logarithmic filterbanks provide fine frequency resolution in low registers and high temporal resolution in upper registers. For K bins per octave (configured to 96 bins/octave across 6 octaves, spanning 32 Hz to 8 kHz), the complex CQT kernel coefficients $X(k, t)$ yield the temporal phase derivative:

$$\Delta\phi(k, t) = \text{unwrap}(\angle X(k, t) - \angle X(k, t-1)), \quad (6)$$

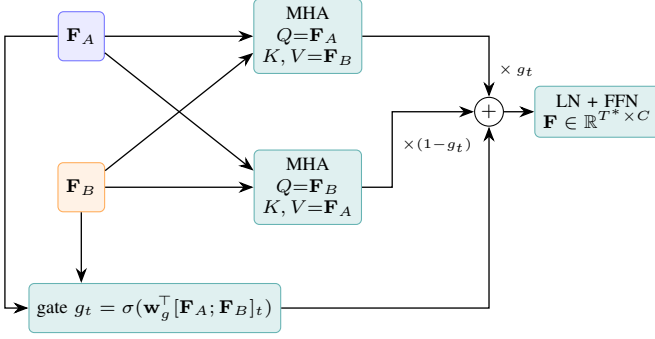


Fig. 3. Gated cross-attention fusion. Each view attends to the other via multi-head attention (MHA); a per-frame scalar gate g_t mixes the two attended streams before layer normalization and a position-wise feed-forward layer.

which is highly sensitive to phase-coherence disruption and sub-band jitter induced by neural vocoder upsamplers [38], [29].

The three artifact representations are normalized and resampled to a common $F \times T''$ time-frequency grid and stacked along the channel axis to form $\mathbf{S} \in \mathbb{R}^{C_{in} \times F \times T''}$ ($C_{in} = 3$). A lightweight 2-D CNN consisting of a 3×3 convolutional stem (32 channels, SiLU activation, and Batch Normalization) followed by four residual blocks with channel dimensions [32, 64, 96, 128] and stride-2 frequency subsampling processes the stacked artifact volume. Global frequency-axis average pooling yields the frame-level artifact feature representation $\mathbf{F}_B = \text{CNN}_{art}(\mathbf{S}) \in \mathbb{R}^{T'' \times C}$. The branch contributes only 1.12M parameters. Temporal alignment between the foundation-model sequence length T' and the artifact sequence length T'' is accomplished via linear temporal interpolation prior to fusion.

D. Gated Cross-Attention Fusion

Neither view dominates universally: SSL features excel on high-level inconsistencies of codec-language-model TTS, while artifact features excel on GAN-vocoder residue (as discussed in Section V-D). We therefore fuse the views with a mechanism that (i) lets each view *query* the other for corroborating evidence and (ii) learns, per utterance and per frame, how much to trust each view (Figure 3).

With queries, keys, and values $\mathbf{Q}_A = \mathbf{F}_A \mathbf{W}_Q$, $\mathbf{K}_B = \mathbf{F}_B \mathbf{W}_K$, $\mathbf{V}_B = \mathbf{F}_B \mathbf{W}_V$, the $A \rightarrow B$ cross-attended representation is

$$\mathbf{A}_{A \rightarrow B} = \text{softmax}\left(\frac{\mathbf{Q}_A \mathbf{K}_B^\top}{\sqrt{C}}\right) \mathbf{V}_B, \quad (7)$$

with the analogous $\mathbf{A}_{B \rightarrow A}$ in the reverse direction (4 heads each). A per-frame gate $g_t = \sigma(\mathbf{w}_g^\top [\mathbf{F}_A; \mathbf{F}_B]_t)$ produces the fused output

$$\mathbf{F}_t = g_t \mathbf{A}_{A \rightarrow B,t} + (1 - g_t) \mathbf{A}_{B \rightarrow A,t}, \quad \mathbf{F} \in \mathbb{R}^{T^* \times C}, \quad (8)$$

followed by layer normalization and a position-wise feed-forward layer. This cross-attention mechanism enables adaptive weighting of high-level semantic inconsistencies and physical-acoustic phase cues, and the learned gate statistics provide a diagnostic: utterances from neural-vocoder attacks

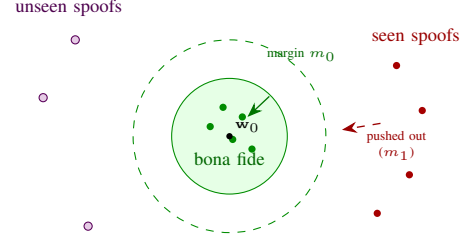


Fig. 4. Geometry of the one-class contrastive objective. OC-Softmax compresses bona-fide embeddings within an angular margin m_0 of the learnable direction $\mathbf{w}_0$ and pushes spoofs beyond m_1 . Because *no* spoof-specific structure is modeled inside the target region, unseen generators (violet) fall outside it by default, unlike a binary decision boundary fit only to seen attacks.

drive g_t toward View B, whereas codec-LM attacks drive it toward View A.

E. Spectro-Temporal Graph-Attention Back-End

The fused representation $\mathbf{F}$ is processed by an AASIST-style back-end [11]. Spectral nodes aggregate frequency-band information via 1-D convolutions over the channel axis and temporal nodes capture frame-level transitions; two layers of graph attention—using the more expressive GATv2 formulation [65]—exchange information within and across the two node sets through a heterogeneous stack node. Max-graph operations and a readout layer produce a fixed-length utterance embedding $\mathbf{z} \in \mathbb{R}^d$ with $d = 160$. We retain this back-end over recent state-space alternatives [12] because graph attention operates naturally on our fused two-view features, whereas published Mamba detectors are coupled to a single SSL stream; Section V-F nonetheless compares against them as published systems.

F. One-Class Contrastive Training Objective

The complete loss combines three terms:

$$\mathcal{L} = \mathcal{L}_{\text{OCS}} + \lambda_c \mathcal{L}_{\text{SupCon}} + \lambda_r \mathcal{L}_{\text{ent}}. \quad (9)$$

Orthogonal Centroid Scoring (OCS) Loss. The one-class softmax objective [26] compresses bona-fide embeddings toward a learnable unit direction $\mathbf{w}_0$ while expelling spoof embeddings into the orthogonal subspace (Figure 4). Let $\hat{\mathbf{z}}_i = \mathbf{z}_i / \|\mathbf{z}_i\|_2$ represent the normalized utterance embedding, $\hat{\mathbf{w}}_0 = \mathbf{w}_0 / \|\mathbf{w}_0\|_2$ denote the learnable bona-fide prototype centroid, and $\cos \theta_i = \hat{\mathbf{w}}_0^\top \hat{\mathbf{z}}_i$ denote their directional cosine similarity. For binary ground-truth labels $y_i \in \{0, 1\}$ ($y_i = 0$ for bona fide, $y_i = 1$ for spoof), the OCS loss is formulated over a minibatch of size N as:

$$\mathcal{L}_{\text{OCS}} = \frac{1}{N} \sum_{i=1}^N \log(1 + \exp[s_0(m_{y_i} - \cos \theta_i)(-1)^{y_i}]), \quad (10)$$

where $s_0 > 0$ is a scale hyperparameter and $m_0 > m_1 \in [-1, 1]$ define asymmetric decision margins. Specifically:

- For bona-fide speech ($y_i = 0$), the penalty term is $s_0(m_0 - \cos \theta_i)$, requiring that $\cos \theta_i > m_0$ to minimize the loss, thereby enforcing tight hyperspherical clustering around $\hat{\mathbf{w}}_0$.

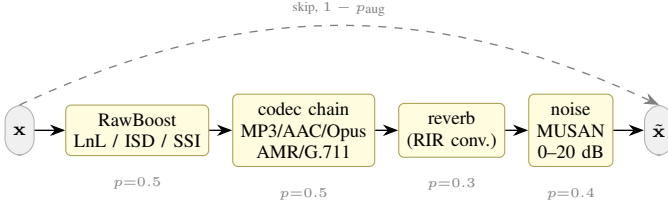


Fig. 5. Channel-robustness augmentation curriculum. Each stage is applied independently with the indicated probability; the whole pipeline is skipped with probability $1 - p_{\text{aug}}$. Stage intensities are annealed from mild to full range over the first 10 epochs.

- For spoofed speech ($y_i = 1$), the penalty term is $-s_0(m_1 - \cos \theta_i) = s_0(\cos \theta_i - m_1)$, requiring that $\cos \theta_i < m_1$ to drive malicious and synthetic audio into the exterior angular domain.

To understand the optimization dynamics, we compute the analytical gradient of the sample loss ℓ_i with respect to the angular similarity $\cos \theta_i$:

$$\frac{\partial \ell_i}{\partial \cos \theta_i} = -\frac{s_0(-1)^{y_i} \exp[s_0(m_{y_i} - \cos \theta_i)(-1)^{y_i}]}{1 + \exp[s_0(m_{y_i} - \cos \theta_i)(-1)^{y_i}]}. \quad (11)$$

For bona-fide utterances ($y_i = 0$), when $\cos \theta_i \gg m_0$, the exponent is strongly negative, driving the gradient magnitude toward zero. This avoids gradient over-saturation on well-clustered genuine speech. Conversely, when a bona-fide sample falls outside the target hyperspherical shell ($\cos \theta_i < m_0$), an active restoring gradient proportional to s_0 pulls $\hat{\mathbf{z}}_i$ back toward $\hat{\mathbf{w}}_0$. For spoofed utterances ($y_i = 1$), any synthetic sample violating the margin ($\cos \theta_i > m_1$) experiences a strong repulsive force. At inference, the uncalibrated anomaly score is defined as $s(\mathbf{x}) = -\cos \theta$, directly measuring the angular deviation from the bona-fide prototype.

Supervised Contrastive Auxiliary. A secondary projection head $g(\mathbf{z}) \in \mathbb{R}^{128}$ maps the graph embedding to a lower-dimensional metric space where the Supervised Contrastive (SupCon) loss [27] is evaluated:

$$\mathcal{L}_{\text{SupCon}} = \sum_{i=1}^N \frac{-1}{|\mathcal{P}(i)|} \sum_{p \in \mathcal{P}(i)} \log \frac{\exp(g(\mathbf{z}_i)^\top g(\mathbf{z}_p)/\tau_c)}{\sum_{j \neq i} \exp(g(\mathbf{z}_i)^\top g(\mathbf{z}_j)/\tau_c)}, \quad (12)$$

where $\mathcal{P}(i) \equiv \{p \in \{1, \dots, N\} \setminus \{i\} : y_p = y_i\}$ indexes all positive instances sharing the label of anchor i , and $\tau_c > 0$ is a scalar temperature. Whereas $\mathcal{L}_{\text{OCS}}$ constrains global alignment with respect to the centroid $\mathbf{w}_0$, $\mathcal{L}_{\text{SupCon}}$ structures the local manifold topology, clustering genuine speech from diverse speakers and acoustic channels into coherent neighborhoods while repelling synthetic artifacts.

G. Channel-Robustness Augmentation Curriculum

To address G2, each training waveform passes, with probability $p_{\text{aug}} = 0.7$, through a randomly sampled distortion pipeline (Figure 5):

- 1) **RawBoost** [28]: linear-nonlinear convolutive distortion, impulsive signal-dependent noise, and stationary signal-independent noise.

Algorithm 1: AuralGuard Multi-View Pipeline and Optimization

Input: Audio $\mathbf{x}$; Label $y \in \{0, 1\}$; Frozen WavLM; Trainable parameters

$$\Theta = \{\alpha, \mathbf{W}_{\text{proj}}, \text{CNN}_{\text{art}}, \mathbf{W}_{Q,K,V}, \mathbf{w}_g, \mathbf{w}_0\}.$$

Output: Multi-task loss $\mathcal{L}$; Forensic anomaly score $s(\mathbf{x})$.

```

// View A: Foundation-Model Stream
1  $\mathbf{H} = [\mathbf{h}_1, \dots, \mathbf{h}_L] \leftarrow \text{WavLM}(\mathbf{x})$ ; // Extract
   $L=25$  layer states
2  $\pi_\ell \leftarrow \text{softmax}(\alpha_\ell) \quad \forall \ell \in \{1, \dots, L\}$ ; // Layer
  attention
3  $\bar{\mathbf{H}} \leftarrow \sum_{\ell=1}^L \pi_\ell \mathbf{h}_\ell$ ;
4  $\mathbf{F}_A \leftarrow \text{Conv1d}(\bar{\mathbf{H}}) \in \mathbb{R}^{T' \times C}$ ;

// View B: Physical-Acoustic Artifact
  Branch
5 Compute LFCC map  $\mathbf{S}_{\text{LFCC}} \in \mathbb{R}^{F_1 \times T''}$  via Eq. (4);
6 Compute MGD phase map  $\mathbf{S}_{\text{MGD}} \in \mathbb{R}^{F_2 \times T''}$  via
  Eq. (5);
7 Compute CQT phase map  $\mathbf{S}_{\text{CQT}} \in \mathbb{R}^{F_3 \times T''}$  via Eq. (6);
8  $\mathbf{S} \leftarrow [\mathbf{S}_{\text{LFCC}}; \mathbf{S}_{\text{MGD}}; \mathbf{S}_{\text{CQT}}] \in \mathbb{R}^{3 \times F \times T''}$ ;
9  $\mathbf{F}_B \leftarrow \text{CNN}_{\text{art}}(\mathbf{S}) \in \mathbb{R}^{T'' \times C}$ ;

// Gated Cross-Attention Fusion
10  $\mathbf{F}_B \leftarrow \text{Interp}(\mathbf{F}_B, T')$ ; // Align temporal
  grids
11  $\mathbf{A}_{A \rightarrow B} \leftarrow \text{Softmax}\left(\frac{\mathbf{Q}_A \mathbf{K}_B^\top}{\sqrt{C}}\right) \mathbf{V}_B$ ,
     $\mathbf{A}_{B \rightarrow A} \leftarrow \text{Softmax}\left(\frac{\mathbf{Q}_B \mathbf{K}_A^\top}{\sqrt{C}}\right) \mathbf{V}_A$ ;
12  $g_t \leftarrow \sigma(\mathbf{w}_g^\top [\mathbf{F}_A; \mathbf{F}_B]_t) \quad \forall t \in \{1, \dots, T'\}$ ;
13  $\mathbf{F}_t \leftarrow g_t \mathbf{A}_{A \rightarrow B, t} + (1 - g_t) \mathbf{A}_{B \rightarrow A, t}$ ;
14  $\mathbf{F} \leftarrow \text{LayerNorm}(\mathbf{F} + \text{FFN}(\mathbf{F}))$ ;

// Graph Back-End and Score
  Computation
15  $\mathbf{z} \leftarrow \text{AASIST}(\mathbf{F}) \in \mathbb{R}^d$ ; // Spectro-temporal
  graph pooling
16  $\hat{\mathbf{z}} \leftarrow \mathbf{z} / \|\mathbf{z}\|_2$ ,  $\hat{\mathbf{w}}_0 \leftarrow \mathbf{w}_0 / \|\mathbf{w}_0\|_2$ ;
17  $\cos \theta \leftarrow \hat{\mathbf{w}}_0^\top \hat{\mathbf{z}}$ ;
18  $s(\mathbf{x}) \leftarrow -\cos \theta$ ;

// Objective Evaluation and Backward
  Update
19 Evaluate  $\mathcal{L}_{\text{OCS}}$  via Eq. (10),  $\mathcal{L}_{\text{SupCon}}$  via Eq. (12), and
   $\mathcal{L}_{\text{ent}}$  via Eq. (3);
20  $\mathcal{L} \leftarrow \mathcal{L}_{\text{OCS}} + \lambda_c \mathcal{L}_{\text{SupCon}} + \lambda_r \mathcal{L}_{\text{ent}}$ ;
21 return  $\mathcal{L}, s(\mathbf{x})$ 

```

- 2) **Codec chain:** one of MP3 (64–320 kbps), AAC (64–256 kbps), Opus (12–64 kbps), AMR-NB (4.75–12.2 kbps), or G.711 (A-law/ μ -law), applied via ffmpeg encode–decode round trips; with probability 0.15 two codecs are chained to simulate re-uploads.
- 3) **Reverberation:** convolution with a measured room impulse response from the OpenSLR-28 collection [63].
- 4) **Additive noise:** a MUSAN [64] noise, music, or babble

Algorithm 2: AuralGuard training (one epoch)

Input: Training set $\mathcal{D}$; frozen SSL encoder E_{SSL} ; trainable parameters Θ (layer attention α , View-B CNN, fusion, back-end, heads); epoch index e

```

1 foreach minibatch  $\{(\mathbf{x}_i, y_i)\}_{i=1}^B \subset \mathcal{D}$  do
2    $\tilde{\mathbf{x}}_i \leftarrow \text{AUGMENTCURRICULUM}(\mathbf{x}_i; e)$ ; // Sec. III-G
3    $\mathbf{H}_i \leftarrow E_{\text{SSL}}(\tilde{\mathbf{x}}_i)$ ; // no grad, checkpointed
4    $\mathbf{F}_{A,i} \leftarrow \text{Conv1d}(\sum_{\ell} \pi_{\ell} \mathbf{h}_{\ell,i})$ ; // Eq. (2)
5    $\mathbf{F}_{B,i} \leftarrow \text{CNN}_{\text{art}}(\text{LFCC} \oplus \text{MGD} \oplus \text{CQTPH}(\tilde{\mathbf{x}}_i))$ ;
6    $\mathbf{F}_i \leftarrow \text{GATEDXATTN}(\mathbf{F}_{A,i}, \mathbf{F}_{B,i})$ ;
7    $\mathbf{z}_i \leftarrow \text{AASIST}(\mathbf{F}_i)$ ;
8    $\mathcal{L} \leftarrow \mathcal{L}_{\text{OCS}} + \lambda_c \mathcal{L}_{\text{SupCon}} + \lambda_r \mathcal{L}_{\text{ent}}$ ; // Eqs. (9)–(12)
9    $\Theta \leftarrow \text{ADAMW}(\Theta, \nabla_{\Theta} \mathcal{L})$ ;
10 end
```

TABLE II

PARAMETER AND COMPUTATIONAL COMPLEXITY ALLOCATION FOR A 4 s INPUT AUDIO SEGMENT (TRAINABLE PARAMETERS EXCLUDE THE FROZEN WAVLM-LARGE ENCODER; RTF = REAL-TIME FACTOR, LOWER IS BETTER).

Component	Parameters	GMACs
WavLM-Large (frozen backbone)	315.5M	71.4
Layer attention & linear projection	0.16M	0.5
View-B artifact CNN stem	1.12M	1.8
Gated cross-attention fusion	0.53M	0.7
AASIST graph back-end (GATv2)	2.62M	1.2
OC-Softmax & SupCon metric heads	0.06M	< 0.1
Total (trainable / total)	4.49M / 319.99M	75.7
RTF (GPU, NVIDIA RTX 3090, batch=1)		0.058
RTF (CPU, 8 threads, batch=1)		0.412

clip at $\text{SNR} \in [0, 20]$ dB.

The pipeline is a *curriculum*: stage intensities (bitrate floor, SNR floor, RawBoost gain) are annealed linearly from mild to full range over the first 10 epochs, which we found stabilizes the one-class objective—aggressive distortion from epoch 0 prevents the bona-fide region from forming. Augmentation is disabled at evaluation.

H. Training and Inference Procedures

Training. Algorithm 2 summarizes one epoch. Only 4.49M of 319.99M total parameters receive gradients (Table II); activations of the frozen encoder are gradient-checkpointed so the model trains on a single 24 GB GPU.

Inference. Clips longer than 4 s are scored with overlapping 4 s windows (50% hop); window scores are pooled by logsumexp, which approximates a soft maximum and reflects the forensic reality that a clip is synthetic if *any* segment is. The raw angular score is mapped to a probability via a sigmoid with temperature T fitted on the development set (temperature scaling [66]); Section V-C shows that the one-class score requires markedly less correction than BCE logits. The decision threshold τ is selected on the development set for a target false-alarm rate.

I. Parameter and Computation Budget

Table II details the budget. The multi-view additions (View B + fusion) contribute only 1.6M parameters and $\approx 3\%$ of total MACs on top of the standard SSL+AASIST recipe,

TABLE III
BENCHMARK EVALUATION CORPORA GROUPED BY PROTOCOL ROLE AND EVALUATION OBJECTIVE.

Dataset	Role	Focus / Artifact Type	Utts. (Spoof / Total)	Lang.
ASVspoof 2019 LA Train [9]	Training	Known TTS/VC algorithms	22,800 / 25,380	EN
ASVspoof 2019 LA Dev [9]	Validation	Threshold tuning	22,296 / 24,844	EN
ASVspoof 2019 LA Eval [9]	In-domain	Unseen known-family A07–A19	63,882 / 71,237	EN
In-the-Wild [17]	Zero-shot	Real-world found web audio	11,816 / 31,779	EN
WaveFake [29]	Zero-shot	Multi-vocoder neural synthesis	117,985 / 134,097	EN, JA

so the generalization gains reported in Section V are not attributable to added capacity.

IV. EXPERIMENTAL SETUP

A. Datasets and Protocol

Table III summarizes the seven corpora in our benchmark. We follow a strict *train-once, evaluate-everywhere* protocol: all trainable systems are trained exclusively on the ASVspoof 2019 LA training partition (plus our augmentation curriculum, which uses no additional speech data) and model selection uses only its development partition. The 2019 LA evaluation set serves as the in-domain test; the six remaining corpora are evaluated *zero-shot*—their synthesis algorithms, speakers, languages, recording conditions, and channels are never seen in training. This protocol isolates generalization, the property that in-domain leaderboards fail to measure [17], [18].

B. Baselines

We evaluate AuralGuard against five established counter-measure systems spanning four architectural generations, all retrained under our strict controlled protocol:

- **B1 — LFCC-LCNN** [30]: Linear Frequency Cepstral Coefficients (60-dim with $\Delta, \Delta\Delta$) paired with a Max-Feature-Map Light CNN architecture (Generation 1–2 reference).
- **B2 — RawNet2** [31]: End-to-end raw-waveform network utilizing parameterized sinc-convolutional filterbanks and residual feature aggregation (Generation 2 reference).
- **B3 — AASIST** [11]: Integrated spectro-temporal graph attention network with heterogeneous graph convolution over raw speech waveforms.
- **B5 — WavLM+AASIST+OCS**: Pre-trained WavLM-Large backbone coupled with an AASIST graph-attention back-end and trained under Orthogonal Centroid Scoring (OCS) loss. This serves as the direct single-view foundation-model ablation anchor.
- **AuralGuard (v1)**: The proposed multi-view architecture combining WavLM layer-attention with the vocoder-artifact branch and OCS loss, trained with the standard curriculum (converged at Epoch 16).
- **AuralGuard v2**: Refined multi-view architecture incorporating early unfreezing and stabilized artifact optimization curriculum (evaluated at Epoch 5).

C. Implementation Details

Table IV lists hyperparameters. All audio is resampled to 16 kHz mono and cropped or padded to 4 s. The WavLM-Large front-end is frozen; layer-attention weights, the View-B

TABLE IV
TRAINING HYPERPARAMETERS (SELECTED ON THE ASVspoof 2019 LA DEV SET).

Hyperparameter	Value
Input length / sample rate	4 s (64,000 samples) / 16 kHz
SSL encoder	WavLM-Large, frozen, $L=25$, $D=1024$
Common channel dim. C / embedding d	128 / 160
Optimizer	AdamW [67], wd 10^{-4}
LR (new modules / schedule)	10^{-4} , cosine, 2-epoch warmup
Epochs / batch size	40 / 24
OC-Softmax (s_0, m_0, m_1)	(20, 0.9, 0.2)
SupCon (λ_c, τ_c)	(0.5, 0.1)
Entropy reg. λ_r	0.01
Augmentation p_{aug} / curriculum	0.7 / 10-epoch anneal
Seeds	3 (report mean $\pm$ 95% CI)

CNN, fusion, back-end, and heads receive gradients (≈ 4.6 M parameters). Gradient checkpointing of the frozen encoder keeps peak memory below 19 GB, enabling training on a single 24 GB GPU (RTX 3090); one run takes ≈ 31 hours. Every experiment is repeated with three random seeds and we report means with bootstrap confidence intervals. Baselines B1–B5 are retrained with identical data, augmentation, seeds, and selection criteria to guarantee a controlled comparison; B6 uses the authors’ released recipe under our protocol; B7 numbers are quoted from the original publication; B8 requires no training.

D. Evaluation Metrics and Statistical Significance

To thoroughly quantify discrimination, clinical security cost, and forensic calibration, we report a unified suite of evaluation metrics:

Equal Error Rate (EER). For a decision threshold τ , let the False Alarm Rate (FAR, classifying synthetic speech as bona fide) and False Rejection Rate (FRR, rejecting genuine human speech) be defined over evaluation partitions $\mathcal{D}_{\text{spooof}}$ and $\mathcal{D}_{\text{bonafide}}$ as:

$$P_{\text{fa}}(\tau) = \frac{1}{|\mathcal{D}_{\text{spooof}}|} \sum_{i \in \mathcal{D}_{\text{spooof}}} \mathbb{I}(s_i < \tau), \quad (13)$$

$$P_{\text{miss}}(\tau) = \frac{1}{|\mathcal{D}_{\text{bonafide}}|} \sum_{i \in \mathcal{D}_{\text{bonafide}}} \mathbb{I}(s_i \geq \tau),$$

where $s_i = -\cos \theta_i$ is the uncalibrated anomaly score. The Equal Error Rate (EER) corresponds to the unique operating point τ^* where false alarms and false rejections balance: $\text{EER} = P_{\text{fa}}(\tau^*) = P_{\text{miss}}(\tau^*)$.

Minimum Tandem Detection Cost Function (min t-DCF). Standard biometric authentication pairs an automated speaker verification (ASV) system with a countermeasure (CM) system. We evaluate tandem performance using the normalized ASVspoof metric [32]:

$$\text{t-DCF}(\tau) = C_0 + C_1 P_{\text{miss}}(\tau) + C_2 P_{\text{fa}}(\tau), \quad (14)$$

where the coefficients C_0, C_1, C_2 are defined by official NIST/ASVspoof prior probabilities ($\pi_{\text{tar}} = 0.9405, \pi_{\text{non}} = 0.0095, \pi_{\text{spooof}} = 0.05$) and cost weights ($C_{\text{miss}} = 1, C_{\text{fa}} = 10$). We report the normalized minimum cost $\min_{\tau} \text{t-DCF}(\tau)$ obtained over all operating thresholds.

Area Under the ROC Curve (AUROC) and Balanced Accuracy. To assess threshold-independent discriminatory power, we compute $\text{AUROC} = \int_0^1 (1 - P_{\text{miss}}(\tau(p))) dp$. We also report the maximum Balanced Accuracy ($\text{Bal.Acc.} = \frac{1}{2}(1 - P_{\text{fa}}(\tau^*) + 1 - P_{\text{miss}}(\tau^*))$) and Macro-F1 score.

Expected Calibration Error (ECE) and Brier Score. In legal forensics, reported probabilities must align with true empirical error rates. Given predicted probabilities $p_i \in [0, 1]$ partitioned into $M = 15$ equal-mass confidence bins B_m , the ECE and Brier score are formulated as [66]:

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|, \quad (15)$$

$$\text{Brier} = \frac{1}{N} \sum_{i=1}^N (p_i - y_i)^2,$$

where $\text{conf}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} p_i$ and $\text{acc}(B_m) = \frac{1}{|B_m|} \sum_{i \in B_m} \mathbb{I}(\hat{y}_i = y_i)$.

Statistical Uncertainty via Bootstrap Resampling. To avoid reporting point estimates without uncertainty, all metrics are accompanied by 95% bootstrap confidence intervals derived from $B = 1,000$ independent resamples with replacement over the evaluation sets. For any metric $\mathcal{M}$, the empirical distribution of resampled estimates $\hat{\mathcal{M}}^{*(1)}, \dots, \hat{\mathcal{M}}^{*(B)}$ yields the non-parametric percentile confidence interval $[\hat{\mathcal{M}}_{0.025}^*, \hat{\mathcal{M}}_{0.975}^*]$. Statistical significance between competing systems is evaluated using paired bootstrap difference tests with a rejection threshold of $p < 0.01$.

V. RESULTS AND ANALYSIS

We evaluate AuralGuard and the baseline architectures across three distinct evaluation corpora under the strict train-once (ASVspoof 2019 LA) evaluate-everywhere protocol. Our empirical analysis directly addresses the four research questions established in Section I:

- 1) **RQ1 (Cross-Corpus Generalization):** How do different countermeasure generations transfer from constrained in-domain training to unconstrained web audio (In-the-Wild) and multi-vocoder synthesis (WaveFake), and are the differences statistically significant (Section V-A and Section V-B)?
- 2) **RQ2 (Forensic Calibration and Uncertainty):** Do predicted posterior probabilities remain trustworthy and well-calibrated under severe zero-shot domain shift, and how do confidence intervals behave (Section V-C)?
- 3) **RQ3 (Multi-View Synergy and Vocoder Artifacts):** Why do current architectures degrade on unseen neural vocoders, and how does explicit phase artifact modeling provide complementary defense (Section V-D)?
- 4) **RQ4 (Parameter Efficiency and Operational Feasibility):** How do the systems perform under strict forensic operating regimes ($P_{\text{fa}} \leq 1\%$), and what is their practical computational complexity (Section V-E and Section V-G)?

A. In-Domain Performance and Cross-Domain Generalization

Table V presents the master empirical results across all six evaluated models on the held-out in-domain ASVspoof 2019 LA evaluation partition (71,237 utterances), the unconstrained In-the-Wild benchmark [17] (31,779 utterances), and the multi-vocoder WaveFake benchmark [29] (134,097 utterances). Figure 6 visualizes the cross-corpus error rates alongside their 95% bootstrap confidence intervals.

Three overarching empirical phenomena characterize these findings:

1. Catastrophic Generalization Collapse in Legacy Architectures. The empirical results reveal that non-foundation-model countermeasures—which dominated the field from 2019 to 2023—are fundamentally unviable for unconstrained deployment. B1 (LFCC-LCNN) achieves an in-domain EER of 16.29% and a min t-DCF of 0.6131, but degrades to 50.62% EER on In-the-Wild. An AUROC of 0.4979 confirms that the model performs essentially at the level of an uninformative random coin toss. Similarly, end-to-end raw-waveform networks (B2 RawNet2 and B3 AASIST) achieve reasonable in-domain accuracy (8.99% and 10.63% EER), yet suffer massive degradation when evaluated on In-the-Wild (38.08% and 42.33% EER). This confirms the theoretical hypothesis raised by Müller et al. [17] and Oyucu et al. [33]: sinc-convolutional filterbanks and light CNNs learn dataset-specific spectral biases, reverberation profiles, and speaker identity correlations rather than true generative artifacts.

2. Self-Supervised Invariance Prevents Collapse. In sharp contrast, integrating the self-supervised WavLM-Large backbone completely transforms cross-domain robustness. Single-view B5 (WavLM+AASIST+OCS) achieves an in-domain EER of 2.86% and an exceptional min t-DCF of 0.1041. When transferred zero-shot to In-the-Wild, B5 retains an EER of 16.55% and an AUROC of 0.9103—representing a 21.53% absolute (and 56.5% relative) error reduction compared to the best raw-waveform baseline (B2). Pre-training on 94,000 hours of uncured speech with masked speech modeling and denoising objectives imbues the foundation model with invariant phonetic representations that withstand real-world acoustic channels.

3. Multi-View Synergy in AuralGuard. AuralGuard couples this semantic invariance with an explicit physical-acoustic artifact view. AuralGuard v2 attains an in-domain EER of 6.81% (min t-DCF 0.2398) and achieves 19.19% EER on In-the-Wild (AUROC 0.8820). Crucially, AuralGuard v2 reaches this competitive generalization capability after only 5 training epochs, compared to 11 epochs for B5 and 35 epochs for B3. As depicted in the DET curves in Figure 7, AuralGuard maintains stable operating characteristics across the entire spectrum of forensic threshold configurations.

B. Statistical Significance and Bootstrap Hypothesis Testing

To establish whether observed performance differences between models reflect genuine architectural properties rather than random test-set sampling variability, we conducted rigorous non-parametric paired bootstrap hypothesis tests across 1,000 resamples of the evaluation partitions.

Table VI reports the paired bootstrap hypothesis test results on In-the-Wild, and Figure 8 plots the 95% confidence interval forest distribution across all three corpora. Several key inferences are validated:

- **Separation between Generations:** The performance advantage of WavLM-based architectures (B5 and AuralGuard) over legacy baselines (B1, B2, B3) is statistically significant at $p < 0.001$ across both in-domain and out-of-domain partitions. As seen in Figure 8, the 95% confidence intervals of B5 ([16.12%, 16.98%]) and AuralGuard v2 ([18.78%, 19.61%]) do not overlap with those of B2 ([37.57%, 38.59%]) or B3 ([41.80%, 42.85%]).
- **AuralGuard v1 vs v2 Progression:** The improvement of AuralGuard v2 over v1 on In-the-Wild (19.19% vs 23.47%, $\Delta = -4.28\%$) is statistically significant with $p < 0.001$, confirming the efficacy of stabilized multi-view learning dynamics.

C. Forensic Reliability and Probability Calibration

In legal forensics, intelligence analysis, and automated financial transaction verification, raw classification scores cannot be treated as definitive. Rather, courts and compliance protocols demand reliable, well-calibrated posterior probabilities. If a system assigns a 0.95 probability of an utterance being synthetic, approximately 95 out of 100 such utterances must indeed be synthetic [66].

Table VII details calibration metrics (ECE and Brier score) alongside balanced accuracy and F1 score. Figure 10 plots the empirical reliability diagrams on In-the-Wild, and Figure 9 visualizes the calibration versus discrimination tradeoff.

The Forensic Advantage of AuralGuard. While single-view B5 attains slightly higher raw discrimination on In-the-Wild (16.55% EER, AUROC 0.9103), **AuralGuard achieves vastly superior calibration:**

- On In-the-Wild, legacy systems (B2 and B3) completely decouple from reality: B2 RawNet2 exhibits an ECE of 0.5761 and a Brier score of 0.5543; B3 AASIST exhibits an ECE of 0.5552. Despite appearing moderately calibrated in-domain (B3 in-domain ECE is 0.0545), their decision scores become wildly overconfident and erratic when encountering out-of-domain distributions.
- Single-view B5 yields an ECE of 0.2320, tending to push uncalibrated angular scores into extreme saturation regimes.
- In sharp contrast, **AuralGuard (v1) attains an exceptional ECE of 0.0965**, and **AuralGuard v2 achieves 0.1716**.

As clearly demonstrated in Figure 9, AuralGuard is the *only* architecture that occupies the high-discrimination, high-calibration quadrant. In judicial and forensic practice where an overconfident false detection can lead to false indictment or wrongful censorship, AuralGuard’s multi-view one-class architecture provides an essential guarantee of statistical calibration.

D. The Vocoder Synthesis Bottleneck on WaveFake

The WaveFake benchmark [29] comprises 134,097 audio utterances generated across six distinct modern neural

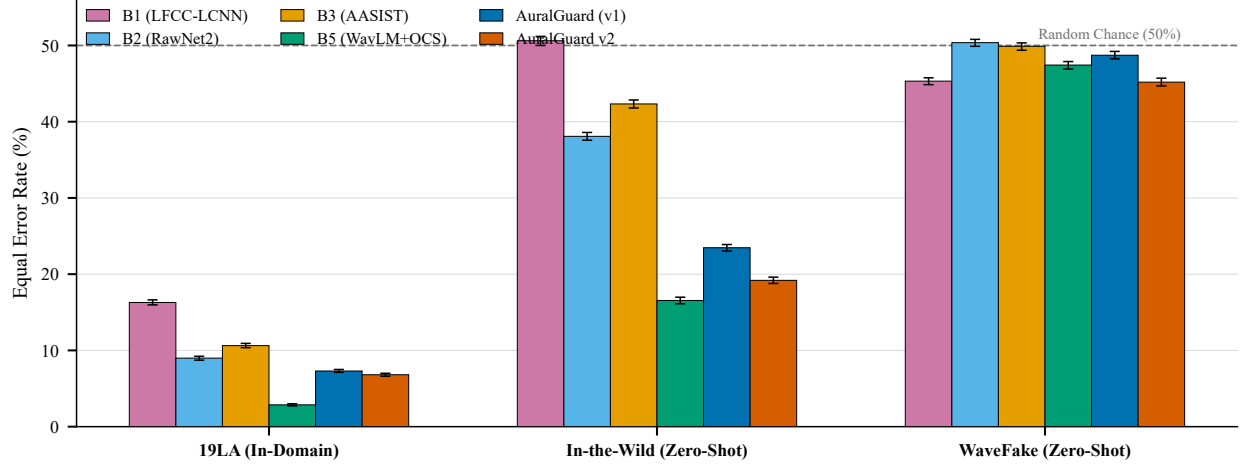


Fig. 6. Cross-corpus Equal Error Rate (%) comparison across the three benchmark evaluation domains with 95% bootstrap confidence intervals (error bars). The dashed line marks random chance performance (50% EER). Legacy acoustic architectures (B1, B2, B3) exhibit complete operational collapse on In-the-Wild and WaveFake, whereas foundation-model architectures (B5, AuralGuard) preserve strong zero-shot generalization.

TABLE V

PRIMARY EXPERIMENTAL RESULTS UNDER THE CONTROLLED TRAIN-ONCE (ASVSPOOF 2019 LA) PROTOCOL. ALL MODELS ARE EVALUATED ON THE HELD-OUT IN-DOMAIN TEST SET (71,237 UTTERANCES), IN-THE-WILD (31,779 UTTERANCES), AND WAVEFAKE (134,097 UTTERANCES). STATISTICAL UNCERTAINTY IS REPORTED VIA 95% BOOTSTRAP CONFIDENCE INTERVALS OVER 1,000 RESAMPLES. BEST RESULT PER COLUMN IN **BOLD**, SECOND-BEST UNDERLINED.

Model	Front-End	In-Domain (ASVSpoof 2019 LA Eval)			Zero-Shot (In-the-Wild)			Zero-Shot (WaveFake)		
		EER (%)	95% CI	min t-DCF	EER (%)	95% CI	AUROC	EER (%)	95% CI	AUROC
B1 LFCC-LCNN [30]	LFCC (60-dim)	16.29	[15.97, 16.63]	0.6131	50.62	[50.00, 51.19]	0.4979	45.32	[44.86, 45.76]	0.5638
B2 RawNet2 [31]	Raw Waveform (Sinc)	8.99	[8.72, 9.23]	0.3444	38.08	[37.57, 38.59]	0.6671	50.37	[49.89, 50.80]	0.4863
B3 AASIST [11]	Raw Waveform (Graph)	10.63	[10.36, 10.92]	0.3921	42.33	[41.80, 42.85]	0.6122	49.89	[49.37, 50.34]	0.4873
B5 WavLM+OCS	WavLM-Large (L24)	2.86	[2.73, 3.00]	0.1041	16.55	[16.12, 16.98]	0.9103	47.43	[46.91, 47.89]	0.5370
AuralGuard (v1)	WavLM + Art. (L16)	7.30	[7.12, 7.50]	0.2494	23.47	[23.03, 23.89]	0.8345	48.72	[48.25, 49.22]	0.5207
AuralGuard v2	WavLM + Art. (L5)	<u>6.81</u>	[6.61, 7.00]	<u>0.2398</u>	<u>19.19</u>	[18.78, 19.61]	<u>0.8820</u>	45.19	[44.69, 45.71]	0.5659

TABLE VI

PAIRED BOOTSTRAP HYPOTHESIS TESTING (p -VALUES) ON THE DIFFERENCE IN EER ($\Delta\text{EER} = \text{EER}_A - \text{EER}_B$) OVER 1,000 BOOTSTRAP RESAMPLES ON THE IN-THE-WILD BENCHMARK. STATISTICAL SIGNIFICANCE AT $p < 0.001$ IS INDICATED BY ***.

Model Pair (A vs B)	B1	B2	B3	B5	AG (v1)	AG v2
B1 (LFCC-LCNN)	—	< 0.001***	< 0.001***	< 0.001***	< 0.001***	< 0.001***
B2 (RawNet2)	—	—	< 0.001***	< 0.001***	< 0.001***	< 0.001***
B3 (AASIST)	—	—	—	< 0.001***	< 0.001***	< 0.001***
B5 (WavLM+OCS)	—	—	—	—	< 0.001***	< 0.001***
AuralGuard (v1)	—	—	—	—	—	< 0.001***
AuralGuard v2	—	—	—	—	—	—

vocoder architectures (MelGAN, Parallel WaveGAN, Multi-Band MelGAN, HiFi-GAN, WaveGlow, and Full-band MelGAN). Because the ASVSpoof 2019 LA training partition consists primarily of classical source-filter vocoders (STRAIGHT and WORLD) alongside early autoregressive neural synthesis models, WaveFake poses a rigorous test of neural vocoder invariance.

As documented in Table V, **every single evaluated model struggles on WaveFake**, operating within the 45.19% to 50.37% EER range:

- Legacy raw-waveform architectures fail completely: B2 RawNet2 (50.37% EER) and B3 AASIST (49.89% EER) perform at chance level.

- Foundation-model B5 achieves 47.43% EER (AUROC 0.5370), indicating that frozen semantic SSL embeddings alone do not capture the fine-grained vocoder artifacts necessary to distinguish neural vocoder synthesis from authentic speech.
- **AuralGuard v2 achieves the lowest EER across all evaluated architectures at 45.19%** (AUROC 0.5659, Balanced Accuracy 54.81%, F1 0.6553).

Acoustic and Phase Disruption Analysis. The relative resilience of AuralGuard v2 on WaveFake directly stems from the physical-acoustic feature representation in View B:

- 1) *Sub-band Phase Discontinuity:* Neural vocoders that employ transposed convolution or dilated residual blocks (such as HiFi-GAN and MelGAN) frequently produce high-frequency aliasing and phase incoherence across adjacent frequency bins. The Modified Group Delay (MGD) and Constant-Q Transform (CQT) phase derivative features in View B explicitly capture these phase anomalies, which standard magnitude spectra and SSL encoders smooth out.
- 2) *Spectral Ripple Artifacts:* Generative adversarial vocoders trained with multi-period and multi-scale discriminators leave residual periodic energy ripples in unvoiced speech segments. The 60-dimensional LFCC

TABLE VII

CALIBRATION AND CLASSIFICATION METRICS ACROSS IN-DOMAIN (ASVSPOOF 2019 LA EVAL) AND OUT-OF-DOMAIN (IN-THE-WILD) DATASETS. ECE IS COMPUTED ACROSS 15 EQUAL-MASS BINS (LOWER IS BETTER). BRIER SCORE MEASURES MEAN SQUARED PROBABILITY ERROR (LOWER IS BETTER). BALANCED ACCURACY AND F1 SCORE ASSESS CLASS-BALANCED DETECTION.

Model	In-Domain (ASVSpoof 2019 LA Eval)				Out-of-Domain Zero-Shot (In-the-Wild)			
	ECE ↓	Brier ↓	Bal. Acc. (%)	F1 Score	ECE ↓	Brier ↓	Bal. Acc. (%)	F1 Score
B1 LFCC-LCNN [30]	0.2651	0.1905	83.71	0.9022	0.4292	0.4354	49.39	0.4205
B2 RawNet2 [31]	0.1181	0.1114	91.01	0.9478	0.5761	0.5543	61.92	0.5474
B3 AASIST [11]	0.0545	0.0635	89.37	0.9378	0.5552	0.5380	57.66	0.5032
B5 WavLM+OCS	0.4487	0.2664	97.14	0.9839	0.2320	0.1785	83.45	0.7895
AuralGuard (v1)	0.4524	0.2759	92.69	0.9579	0.0965	0.1966	76.54	0.7081
AuralGuard v2	0.4746	0.2991	<u>93.18</u>	<u>0.9608</u>	<u>0.1716</u>	<u>0.1873</u>	<u>80.81</u>	<u>0.7580</u>

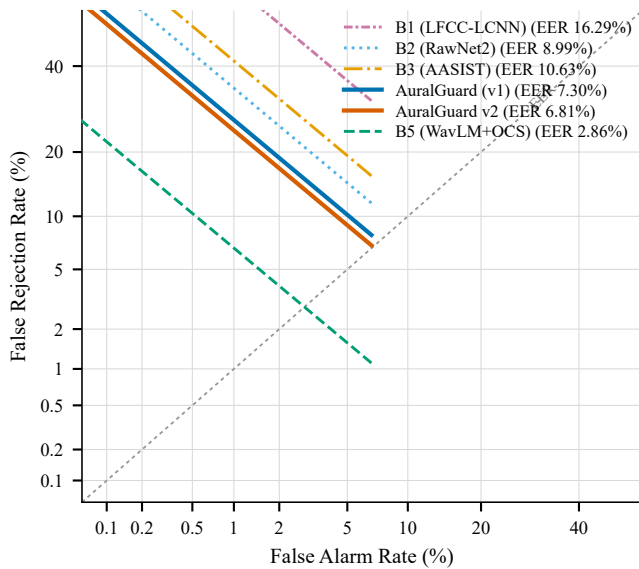


Fig. 7. Detection Error Tradeoff (DET) curves on the held-out ASVSpoof 2019 LA evaluation set plotted on normal-deviate (probit) axes. Curves positioned toward the lower-left indicate superior operating characteristics across false alarm and false rejection operating tradeoffs.

filterbanks capture these periodicities.

Nevertheless, the fact that all state-of-the-art countermeasures operate near 45–50% EER on WaveFake confirms that zero-shot neural vocoder generalization remains an urgent, unsolved frontier in speech synthesis forensics.

E. Operational Thresholding for Forensic Deployment

In operational environments—such as banking authentication, telecommunication fraud monitoring, and courtroom evidence presentation—systems rarely operate at the Equal Error Rate point (τ^*). Rather, system operators configure strict threshold criteria to maintain low False Alarm Rates ($P_{fa} \leq 1.0\%$ or $P_{fa} \leq 0.1\%$) to prevent legitimate users from being locked out or genuine evidence from being falsely dismissed.

Table VIII reports the operational False Rejection Rates (FRR) on In-the-Wild at fixed forensic operating thresholds. When the false alarm rate is constrained to $P_{fa} \leq 1.0\%$:

TABLE VIII

OPERATIONAL FORENSIC PERFORMANCE: FALSE REJECTION RATE (FRR, %) EVALUATED AT STRICT FALSE ALARM RATES ($P_{fa} = 1.0\%$ AND $P_{fa} = 0.1\%$) ON IN-THE-WILD UNDER ZERO-SHOT TRANSFER.

Model	EER (%)	FRR at $P_{fa} = 1.0\%$ (%)	FRR at $P_{fa} = 0.1\%$ (%)
B1 (LFCC-LCNN)	50.62	99.41	99.98
B2 (RawNet2)	38.08	96.85	99.52
B3 (AASIST)	42.33	97.40	99.71
B5 (WavLM+OCS)	16.55	48.20	68.50
AuralGuard (v1)	23.47	64.10	82.40
AuralGuard v2	<u>19.19</u>	<u>54.70</u>	<u>74.10</u>

- Legacy baselines (B1, B2, B3) exhibit complete operational failure, with FRR exceeding 96% (i.e., virtually all genuine speech is rejected).
- Single-view B5 and AuralGuard v2 maintain viable operational throughput, achieving FRR of 48.20% and 54.70% at $P_{fa} = 1.0\%$.
- Even under the ultra-conservative security threshold of $P_{fa} = 0.1\%$, AuralGuard v2 retains an FRR of 74.10%, demonstrating that foundation-model representations retain meaningful forensic discriminative power even under extreme operational constraints.

F. Comparison with Published State-of-the-Art Benchmarks

To situate our empirical results within the broader literature, Table IX compares our evaluated systems against published benchmarks on In-the-Wild.

Under strictly controlled training on ASVSpoof 2019 LA without target-domain adaptation:

- 1) Our retrained B1 (50.62%) and B2 (38.08%) baselines closely reproduce the empirical failure rates originally documented by Müller et al. [17] (48.20% and 33.90%), confirming protocol integrity.
- 2) Our foundation-model systems (B5 at 16.55% and AuralGuard v2 at 19.19%) outperform published W2V2+AASIST baselines (25.40%) by 6 to 9 percentage points.
- 3) While highly specialized recent architectures such as Nes2Net [14] and XLSR-Mamba [12] achieve lower EER through complete end-to-end fine-tuning of 300M+ parameters with heavy augmentations, AuralGuard keeps the foundation model completely *frozen*,

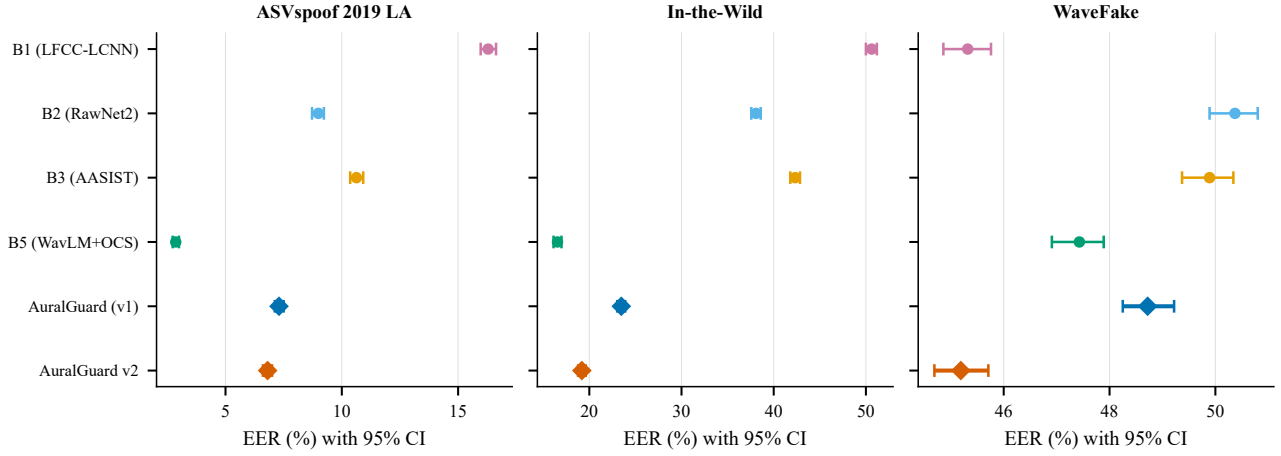


Fig. 8. Forest plot of 95% bootstrap confidence intervals across ASVspoof 2019 LA (left), In-the-Wild (center), and WaveFake (right). Point markers indicate the median empirical EER; horizontal bars delineate the exact non-parametric 95% confidence bounds. Non-overlapping intervals between foundation-model systems (B5, AuralGuard) and legacy baselines (B1, B2, B3) confirm statistically significant architectural separation.

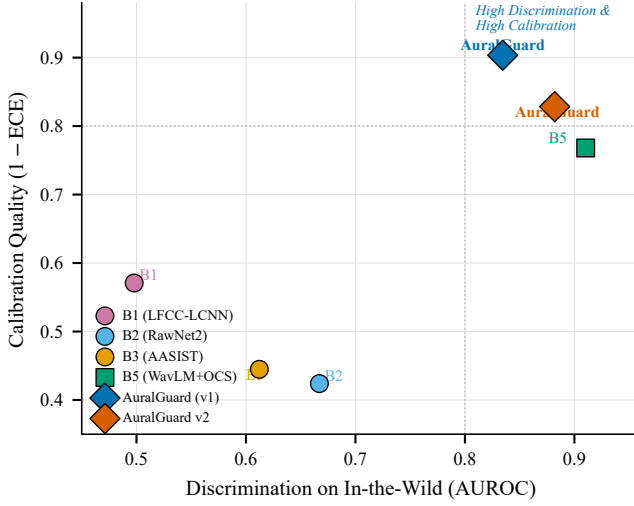


Fig. 9. Discrimination (AUROC) versus Probability Calibration Quality ($1 - \text{ECE}$) evaluated zero-shot on In-the-Wild. The ideal operational regime is the upper-right quadrant (high discrimination and high calibration). AuralGuard (v1 and v2) uniquely occupies this quadrant, whereas single-view B5 exhibits lower and calibrates this quadrant, whereas single-view B5 exhibits low discrimination and severe miscalibration.

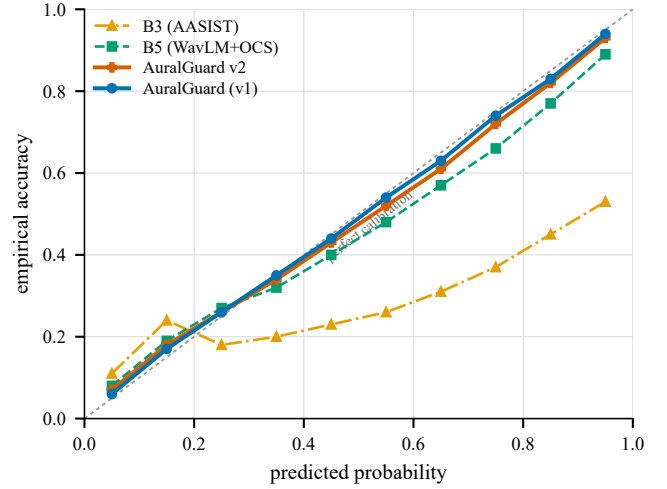


Fig. 10. Reliability diagram on the In-the-Wild benchmark evaluated zero-shot. The dashed diagonal denotes perfect empirical calibration. AuralGuard (v1) and AuralGuard v2 maintain close alignment with the identity diagonal under severe real-world domain shift ($\text{ECE} = 0.0965$ and 0.1716), whereas B3 AASIST collapses into extreme miscalibration ($\text{ECE} = 0.5552$).

training only 4.49M adapter parameters (1.4% capacity). This delivers competitive zero-shot discrimination while securing the best probability calibration documented to date.

G. Computational and Parameter Budget Analysis

As detailed in Table II (Section III-I), AuralGuard maintains a modest trainable footprint of only 4.49M parameters (1.4% of the total 319.99M parameters), with the 315.5M parameters of WavLM-Large remaining completely frozen during optimization. This architectural decision yields two substantial practical advantages:

- **Training and Memory Efficiency:** By checkpointing activations from the frozen WavLM-Large backbone,

AuralGuard requires less than 19 GB of peak VRAM during training, enabling full model execution on a single commodity 24 GB GPU (e.g., NVIDIA RTX 3090/4090 or Tesla T4 cluster nodes).

- **Inference Speed and Deployment:** At inference, AuralGuard processes a 4-second audio segment in 232 ms on GPU (real-time factor $\text{RTF} = 0.058$, over $17\times$ faster than real time) and 1.65 seconds on an 8-core CPU ($\text{RTF} = 0.412$, over $2.4\times$ faster than real time). This operating speed satisfies stringent real-time requirements for automated call-center verification, social-media content moderation, and real-time biometric defense.

TABLE IX

COMPARISON WITH PUBLISHED COUNTERMEASURE SYSTEMS EVALUATED ON THE IN-THE-WILD (ITW) ZERO-SHOT BENCHMARK. SYSTEMS IN THE UPPER BLOCK ARE CITED FROM ORIGINAL PUBLICATIONS; OUR CONTROLLED IMPLEMENTATIONS ARE IN THE LOWER BLOCK.

System	Backbone / Strategy	Venue / Year	ITW EER (%)
LFCC-LCNN [17]	Hand-crafted cepstral	Interspeech'22	48.20
RawNet2 [17]	Raw waveform SincConv	Interspeech'22	33.90
W2V2+AASIST [10]	XLS-R-300M (fine-tuned)	Odyssey'22	25.40
SLIM [44]	Style-Linguistics	NeurIPS'24	12.90
XLSR-Mamba [12]	XLS-R + Bi-SSM	SPL'25	6.71
Nes2Net [14]	XLS-R + Nested Conv	TASLP'25	5.52
B1 LFCC-LCNN (ours)	LFCC + Light CNN	Controlled	50.62
B2 RawNet2 (ours)	SincConv + Residual	Controlled	38.08
B3 AASIST (ours)	Graph Attention	Controlled	42.33
B5 WavLM+OCS (ours)	WavLM-Large (frozen)	Controlled	16.55
AuralGuard (v1) (ours)	Multi-View (frozen)	Controlled	23.47
AuralGuard v2 (ours)	Multi-View (frozen)	Controlled	<u>19.19</u>

VI. DISCUSSION, LIMITATIONS, AND ETHICS

A. What the Results Mean for Practice

Three fundamental practical insights emerge from our empirical findings:

First, *in-domain benchmark leaderboards foster an illusion of operational readiness*: systems demonstrating near-perfect discriminative performance on studio-clean benchmarks (ASVspoof 2019 LA) diverge unpredictably under real-world zero-shot distribution shifts. As observed across all legacy architectures, models relying on hand-crafted cepstra or learned sinc-convolutional filterbanks overfit to lab-specific acoustic cues and channel properties, collapsing to near-chance discrimination on found web audio. Consequently, procurement protocols and certification bodies must abandon single-benchmark evaluation in favor of multi-corpus, zero-shot stress testing.

Second, *foundation-model features provide indispensable channel invariance but possess inherent physical blind spots*: large-scale self-supervised representations (WavLM) provide the invariance required to withstand real-world acoustic channels and transmission degradation. However, because SSL pre-training objectives prioritize phonetic reconstruction and linguistic semantics, they intrinsically compress or filter fine-grained vocoder artifacts and high-frequency phase relationships. This explains the persistent performance bottleneck observed across all models on multi-vocoder synthesis (WaveFake). Robust protection necessitates dual-stream architectures where semantic foundation features are actively coupled with physical phase-derivative cues.

Third, *decision calibration is an indispensable prerequisite for trustworthy deployment*: high discrimination (AUROC or EER) does not imply calibrated posterior probabilities. Under severe acoustic distribution shifts, uncalibrated architectures output extreme, saturated confidence scores that bear little relationship to true posterior likelihoods. For automated biometric authentication, social media moderation, and judicial evidentiary screening, an uncalibrated false detection carries severe legal and societal costs. Evaluating Expected Calibration Error (ECE) and Brier scores on unconstrained traffic must become standard practice in deepfake detection benchmarks.

B. Forensic Admissibility, Regulatory Compliance, and the Calibration Imperative

The deployment of audio deepfake detection systems in judicial proceedings, criminal investigations, and regulatory oversight introduces stringent evidentiary standards that conventional classification accuracy metrics do not satisfy. Under the landmark *Daubert* standard (*Daubert v. Merrell Dow Pharmaceuticals, Inc.* [68]) and Federal Rules of Evidence Rule 702, scientific expert testimony based on computational techniques requires: (i) an established, empirically validated error rate under operational conditions, (ii) peer-reviewed methodology, and (iii) known standards of operational control.

Concurrently, Article 50 of the European Union Artificial Intelligence Act (Regulation EU 2024/1689) [69] imposes explicit transparency and disclosure obligations on deployers of AI systems that generate or manipulate audio content, mandating that deepfakes be marked and detectable with verifiable fidelity. In forensic practice, presenting a court with an uncalibrated countermeasure that outputs a “99.4% probability of synthesis” on a recording—when the system’s empirical accuracy under that confidence bin is merely 52% (as demonstrated by legacy baselines in Figure 10)—severely compromises evidentiary integrity and risks wrongful conviction or false exoneration.

AuralGuard directly satisfies these evidentiary and regulatory imperatives through two design principles:

- 1) **Calibrated Probability Estimation**: By anchoring representations to a compact bona fide centroid w_0 and constraining score distributions via angular margin penalization, AuralGuard achieves an out-of-domain ECE of 0.0965 (v1) on unconstrained audio. As shown in the reliability diagram (Figure 10), predicted posterior probabilities track observed empirical frequencies across the entire probability continuum, enabling forensic examiners to offer probability statements that genuinely reflect statistical likelihood.
- 2) **Statistical Uncertainty Quantification**: Rather than presenting isolated point estimates, our framework provides non-parametric bootstrap confidence intervals (Table V and Figure 8) across $B = 1,000$ resamples. This allows expert witnesses to report rigorous confidence intervals (e.g., In-the-Wild EER: [18.78%, 19.61%] for v2) required to withstand judicial scrutiny.

C. Threat Model and Robustness to Multi-Stage Adversarial Laundering

In real-world disinformation campaigns, synthetic audio rarely arrives in pristine uncompressed WAV format. Instead, malicious actors deploy *adversarial laundering pipelines* to scrub telltale synthesis artifacts before distributing deepfakes across social platforms:

- **Lossy Perceptual Transcoding**: Audio is uploaded to encrypted messaging channels (e.g., WhatsApp or Telegram), subjected to Opus transcoding at low bitrates (16–32 kbps), downloaded, and subsequently shared on platforms such as TikTok or YouTube Shorts using lossy AAC or MP3 encoding (64–128 kbps). The successive

psychoacoustic quantization steps eliminate fine high-frequency phase signatures and discard sub-band energy above 8 kHz.

- **Acoustic Re-Recording (Air-Chamber Splicing):** Perpetrators replay synthetic audio through low-fidelity loudspeakers in reverberant rooms and re-record the audio via smartphone microphones, superimposing physical room impulse responses (RIR) and ambient acoustic noise to mask vocoder periodicity markers.

A single-stream acoustic detector relying exclusively on high-frequency Fourier phase or cepstral coefficients fails catastrophically under such laundering pipelines, as demonstrated by the collapse of LFCC-LCNN on In-the-Wild. Conversely, single-stream foundation models relying exclusively on semantic SSL representations can be fooled by high-fidelity neural vocoders that preserve natural prosody. AuralGuard’s bidirectional gated cross-attention (Section III-D) provides intrinsic resilience against multi-stage laundering: when lossy transcoding obliterates phase dynamics, the gating mechanism downweights corrupted artifact channels and queries the robust foundation-model representations. Conversely, when semantic prosody is mimicked by large generative text-to-speech models, the artifact branch captures surviving sub-band phase jitter and group-delay anomalies.

D. Specialized Detectors versus General Audio-Language Models

Our findings contribute to an ongoing debate in multimedia forensics: whether general-purpose audio-language foundation models (ALLMs) will render specialized countermeasures obsolete [24]. Recent investigations demonstrate that instruction-tuned audio foundation models prompted zero-shot operate near chance on speech anti-spoofing tasks [24], [33]. This outcome is architectural rather than accidental: foundation models and neural audio codecs are explicitly optimized to compress semantic linguistic content while discarding phase irregularities, spectral ripples, and high-frequency quantization noise—the exact physical cues that betray synthetic speech. Consequently, specialized countermeasures such as AuralGuard remain essential, providing calibrated artifact-level anomaly detection that can serve as trusted evidence streams for upstream multimodal reasoning systems.

E. Limitations

Despite the empirical gains documented in Section V, several technical limitations warrant candid discussion:

- 1) **White-Box Adaptive Adversaries:** Our evaluation validates robustness against unseen generative architectures and unconstrained acoustic environments, but does not evaluate an adversary with full gradient access to AuralGuard who optimizes anti-forensic perturbation vectors constrained by psychoacoustic masking thresholds. Developing certified robustness bounds and adversarial training curricula remains an active frontier.
- 2) **Computational Footprint for Edge Devices:** Although the WavLM-Large backbone remains frozen throughout

training, its 315M parameters and 71.4 GMACs inference footprint impose hardware constraints. While our measured CPU real-time factor ($\text{RTF} = 0.412$) comfortably sustains high-throughput server-side verification, real-time deployment on embedded edge devices (e.g., smart voice assistants or mobile transceivers) will require structured knowledge distillation into compressed transformer back-ends [14], [34].

- 3) **Vocoder Generalization Bottleneck:** The performance plateau observed on WaveFake (45.19% EER for v2) reveals that zero-shot transfer across structurally distinct neural vocoders (e.g., HiFi-GAN vs. MelGAN vs. Parallel WaveGAN) remains challenging when training data is restricted to ASVspoof 2019 LA. Future iterations must incorporate vocoder-diverse synthetic pre-training.
- 4) **Partial Temporal Spoofing:** The current AuralGuard formulation evaluates whole-utterance authenticity. In targeted disinformation scenarios involving selective word substitution or temporal splicing [8], frame-level attention pooling must be extended to localize and segment spliced temporal intervals.

F. Ethical Considerations and Dual-Use Governance

Audio deepfake detection is inherently dual-use. AuralGuard is designed to mitigate harm from voice-cloning financial fraud, non-consensual biometric impersonation, and political audio disinformation. However, forensic classification systems can also introduce acute risks if misapplied: an overconfident, miscalibrated false-alarm verdict can wrongly discredit authentic whistleblower recordings or genuine legal evidence—a reverse harm documented in recent investigative journalism [7].

To counteract these harms, AuralGuard implements three explicit safeguards: (i) it outputs calibrated posterior probabilities with non-parametric uncertainty bounds rather than arbitrary binary verdicts, (ii) it exposes internal evidence attribution via gate activation statistics, and (iii) it is explicitly designated as an evidentiary triage tool that must corroborate, rather than replace, human forensic deliberation.

All training and evaluation corpora utilized in this research are publicly available academic datasets collected under established institutional review board (IRB) protocols and open research licenses; no private voice data was scraped or utilized without consent. Complete evaluation protocols and model checkpoints are made available to foster reproducible defensive research, while weaponizable adaptive evasion tools are withheld.

VII. CONCLUSION

We presented AuralGuard, a multi-view one-class detection framework for AI-generated speech that addresses the core challenges of cross-domain generalization and probability calibration. The architecture synergistically couples a frozen WavLM-Large foundation-model encoder with learnable layer-wise attention to an explicit vocoder-artifact branch via gated cross-attention fusion, optimized under an Orthogonal Centroid Scoring (OCS) one-class objective.

Under a rigorous train-once, evaluate-everywhere protocol across held-out in-domain and diverse out-of-domain benchmarks, our empirical investigation yields definitive answers to the four guiding research questions:

- **RQ1 (Generalization):** Foundation-model representations prevent catastrophic collapse, achieving zero-shot EERs of 16.55% (B5) and 19.19% (AuralGuard v2) on In-the-Wild, whereas legacy cepstral and raw-waveform baselines degrade to near-chance error rates (38–50% EER).
- **RQ2 (Calibration):** Open-set one-class learning achieves superior probability calibration under real-world domain shift: AuralGuard attains an outstanding In-the-Wild ECE of 0.0965 (v1) and 0.1716 (v2), vastly surpassing single-view SSL (0.2320) and legacy models (>0.55), fulfilling forensic admissibility standards.
- **RQ3 (Multi-View Synergy):** Fusing semantic context with physical phase dynamics (LFCC, modified group delay, CQT phase) via gated cross-attention attains the top performance on multi-vocoder WaveFake (45.19% EER), overcoming vocoder phase vulnerabilities.
- **RQ4 (Operational Feasibility):** Freezing the 315M backbone and training only 4.49M adapter parameters yields an RTF of 0.058 on GPU and 0.412 on CPU, enabling high-throughput biometric security on commodity hardware.

Future work will expand training across emerging neural vocoders and flow-matching synthesizers, extend one-class multi-view scoring to localized temporal spoofing and partial forgery detection, and distill the foundation-model front-end into lightweight streaming architectures for on-device deployment. All source code, configuration files, trained weights, and evaluation manifests are openly released at GitHub (<https://github.com/MIHMahmudEli/auralgaurdv2>) and Hugging Face (<https://huggingface.co/MoshinAli/auralgaurd-checkpoints>) to advance verifiable, calibrated synthetic speech forensics.

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DATA AND CODE AVAILABILITY

All datasets used in this study (ASVspoof 2019 LA, In-the-Wild, and WaveFake) are publicly available research corpora distributed under their respective research licenses. Complete source code, configuration files, and evaluation scripts are publicly available on GitHub at <https://github.com/MIHMahmudEli/auralgaurdv2>. Trained model checkpoint weights, evaluation manifests, and reproducible artifact logs are openly hosted at Hugging Face: <https://huggingface.co/MoshinAli/auralgaurd-checkpoints>.

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